American Computer Science League

2023-2024 ● Contest 3: Short Problem Solutions ● Elementary Division

1. Boolean Algebra

Whole numbers less than 20 are the set of numbers {0, 1, 2, 3, 4, …, 19, 20}. The expression $(x < 3)$ requires the value for x to be less than 3 which is {0, 1, 2}. The value for x may also be greater than or equal to 8 and less than or equal to 15 which is the set of {8, 9, 10, 11, 12, 13, 14, 15}. The total number of values in either set is 11.

D. 11

2. Boolean Algebra

The expression $\sim(A + \sim B) + \sim(\sim A + B)$ can be evaluated by simplifying or using a Truth Table. Using DeMorgan's Law, the simplified expression is: $\sim A * \sim\sim B + \sim\sim A * \sim B = \sim AB + A\sim B$
The ordered pairs that work are (1,0) and (0,1).

C. (0,1), (1,0)

3. Boolean Algebra

The following Truth Table has each column filled in:

A	B	$A + B$	AB	$\sim(AB)$	$\sim(A + B)$	$\sim(AB) + \sim(A + B)$
0	0	0	0	1	1	1
0	1	1	0	1	0	1
1	0	1	0	1	0	1
1	1	1	1	0	0	0

Therefore, the solution from top to bottom in the last column is: 1110

A. 1110

4. Boolean Algebra

Simplifying $AB(\sim AB + \sim B + AB)$
$= AB\sim AB + AB\sim B + ABAB$
$= 0 + 0 + AB$
$= AB$

B. AB

5. Boolean Algebra The expression $\sim(AB) + \sim(A + B)$ can be simplified to: $\sim A + \sim B + \sim A \sim B$ $\sim A$ is true for (0,1), (0,0). $\sim B$ is true for (0,0), (1,0). $\sim A \sim B$ is true for (0,0). Choices $\sim A + \sim B$ and $\sim(A * B)$ are equivalent to each other and each is true for everything but (1,1).	C. a, d